Is Recourse Equally Open to Everyone?

Auditing algorithmic recourse in eligibility-style decisions: the 1994 benchmark vs 2023 North Carolina census data

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Code, app & full report

1 Explanations should show what you can do

Counterfactual explanations, a core XAI method, answer: what would need to change for the model to say yes? If a person can act on the answer, it is **recourse**.

DENIED	→	APPROVED
works 40 h/week		works 45 h/week

An explanation says why you were denied. Recourse says what you can do.“Be older” or “get married” may flip the model, but no one can act on that.

2 The question

When tools suggest how to reverse an automated denial, is the path realistic, and equally open to every group? Do findings from the 1994 benchmark still hold in 2023?

3 We compared realistic and off-the-shelf recourse

Data	1994 US Census (Adult) and 2023 North Carolina census (ACS), with the same inflation-adjusted income bar “Approved” = income above $50K in 1994, $103K in 2023 (the same bar in 2023 dollars).
Models	Gradient boosting and logistic regression. Sex and race are not model inputs
Realistic	Only what a person can change: more school, hours, job. Never age, marriage, or birthplace
Off-the-shelf	Anything may change, as generic counterfactual tools allow
Evidence	~800 denied applicants per dataset; 95% bootstrap CIs; groups compared at the same starting distance

Under-30s have almost no realistic path to approval

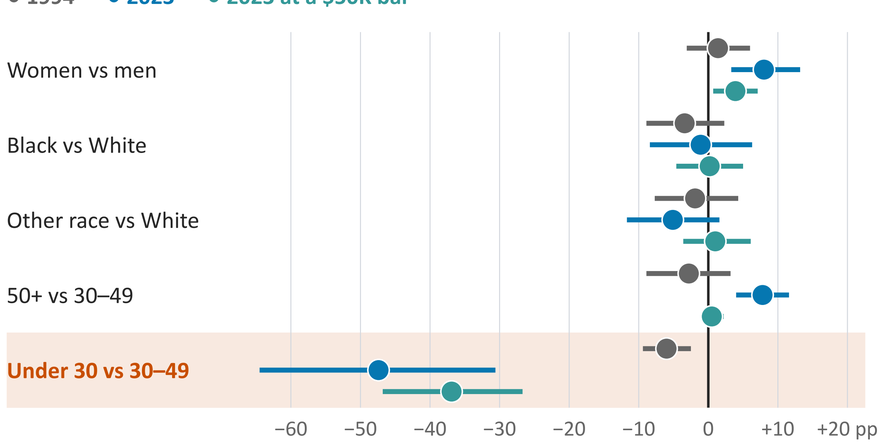
They were 6 percentage points less likely than 30–49-year-olds to have one in 1994, and 47 points less likely in 2023.

98% of qualified under-30s are denied (2023)

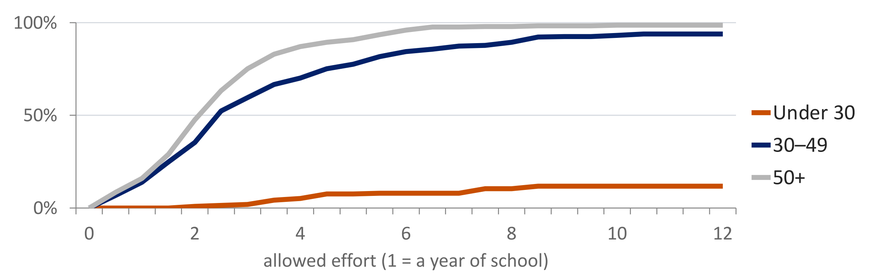
8% of denied under-30s have a realistic path

84% of denied 30–49-year-olds do

Recourse gap vs the largest group, at the same starting distance (95% CI)



More effort does not close the gap (2023)

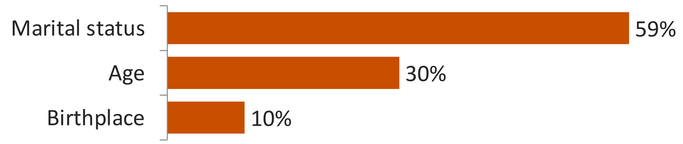


4 Off-the-shelf explanations mostly advise the impossible

77% of its advice breaks a rule (1994, capital gains fixed)

42% still in 2023

What it tells people to change (1994, capital gains fixed)



93% of its advice to denied under-30s in 2023 amounts to “be older” (median: be 4

5 1994’s sex and race gaps came from where people start

1994	Raw gap	Same start
Women vs men	**−19 pp***	+1 pp
Black vs White	**−21 pp***	−3 pp

Treated alike at the same distance from approval. * 95% CI excludes 0

6 Audit explanations, not just accuracy

- Error rates show under-30s are wrongly denied; recourse shows they can’t do anything about it, once denied.
- Generic explanation tools mislead most where real recourse is scarcest.
- Re-audit on current data: 1994 understated the age gap.

7 Limitations

- Income is a proxy label: in real benefit programs, lower income qualifies.
- Costs and what counts as “immutable” are judgment calls.

References: Wachter et al. 2017; Ustun et al. 2019; Gupta et al. 2019; Mothilal et al. 2020; Ding et al. 2021; Karimi et al. 2022 (full list in REPORT.md). Research audit on public survey data, not an eligibility system.